

These are all *normative* statements, or statements about what investors should do (see chapter 2 for normative vs. positive economics). In reality, many do exactly the opposite. A few traders like to hold very risky assets in their personal portfolios, despite their labor income being highly dependent on the stock market. All too many investors hold outsized positions in their employers' stock, and they hold *home-biased* portfolios with little international exposure. These investors load up on the same risks in their financial portfolios as their human capital is exposed to and fail to diversify away some of their human capital risk (see also chapter 4).

2.4. LABOR INCOME RISK

Measuring labor income risk is tricky; we can't observe the value of human capital because it is not traded.

The first economist to treat human capital as an asset and put a value on it was Gary Becker (1964), who won the Nobel Prize in 1992. Becker showed that education is an investment in human capital, rather than just a luxury or consumption expenditure. He also showed that investment in human capital was integral to economic growth. A large literature in labor economics measures human capital returns by computing the benefits to investments in human capital: labor economists compute extra wages earned by those who have gone to graduate school over those who have gone to college, over those with only high school educations, and over those who did not attend high school. Not surprisingly, wages increase with investment in education going from less than a high school diploma (median earnings of \$471 per week in 2012 according to the Bureau of Labor Statistics) to bachelor degrees (\$1,066) to professional degrees (\$1,735). Interestingly, there is a drop to \$1,624 for PhDs. (Someone should have told me before I sweated over mine.) As is well known, women earn less than men, and blacks earn less than whites.

Human capital is the present value of all future wage flows. The return on human capital is often computed as the growth in wages.¹³ Palacios-Huerta

¹³This is assumed by Shiller (1995), Jagannathan and Wang (1996), and Eiling (2013), among others. Denoting wages as L_t and the growth rate of wages as g_H , suppose wages follow the process:

$$L_t = (1 + g)L_{t-1} + \varepsilon_t,$$

where wages start at time 0 from the level L_0 and ε_t is an independent and identically distributed (i.i.d.) shock. If we discount the wage stream at the expected rate of return on human capital, r_H , then we can value the present value of wages as

$$H_t = \frac{L_t}{r_H - g},$$

and the growth in wages, g , is equal to the return on human capital, $H_t/H_{t-1} - 1$. Earlier work by Friedman (1957) and Hall (1978) discounted labor income at risk-free rates to compute human capital.